

Domain Knowledge Chatbot:
A Generative AI Fine-Tuning Project

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Milestone 1: Proposal and Data Selection

Abstract

This proposal describes the Domain Knowledge Chatbot, a system that uses a fine-tuned AI model to help software developers find information about their codebase quickly. The system pulls data from sources like Git histories, design documents, and API specifications. It uses a Retrieval-Augmented Generation (RAG) approach so answers are grounded in real project documents. The goal is to reduce time spent searching for information and make onboarding easier for new team members.

Keywords: generative AI, fine-tuning, RAG, knowledge management, large language models

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When developers move between teams or encounter unfamiliar code, a lot of time is lost trying to find answers. Documentation is often outdated or hard to search. This project aims to solve that problem by building the Domain Knowledge Chatbot — an AI system that can answer plain-language questions about a specific codebase and project domain.

Data Source Quality

The system will use data from Git commit histories, source code files, design documents, API specifications, and knowledge transfer documents. These sources together cover both the technical details and the reasoning behind how the system was built.

Before the data is used, it will be cleaned to remove duplicates and fix inconsistencies. The system will also update automatically as the codebase changes, so the information stays current.

Research Question Quality

The main research question is: Can a fine-tuned language model with a RAG framework give accurate and useful answers to developer questions about a specific project, better than a general-purpose model?

This will be tested by comparing the system's answers to correct answers written by team members, and by collecting user satisfaction ratings. Open-source models like Llama-2 or Mistral will be used as the starting point, keeping the project within practical computing limits.

Proposal Grammar

This proposal uses clear and straightforward language. Technical terms are explained where needed. All citations follow APA 7th edition style, with an author-date format in the text and a full reference list at the end.

Proposal Organization

The proposal is organized into five sections: data sources, research question, grammar, organization, and ethics. This order moves from what data will be used, to what will be studied, to how the document itself is written, and finally to responsible use. Future milestones will cover model training, testing, and results.

Ethical Considerations

Several ethical issues need to be considered. First, Git histories include personal information like developer names and emails. This data will be anonymized where appropriate and handled according to privacy rules such as GDPR.

Second, AI models can sometimes give wrong answers that sound correct. To reduce this risk, every response will be tied to actual source documents so users can verify the information themselves.

Third, the system should support developers, not replace their own thinking. The interface will encourage users to explore the source material rather than just accept AI-generated answers. Finally, areas of the codebase with little documentation may produce weaker results, and this limitation will be clearly communicated to users.

References

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